Fiscal Year:	FY 2016	**Task Last Updated:**	FY 08/01/2016
PI Name:	Wang, Huichen		
Project Title:	Molecular Basis of DNA Repair and Protection from Apoptosis in Neuronal Progenitors Exposed to Space Radiation		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Prairie View	**State:**	TX
Zip Code:	77446	**Congressional District:**	10
Comments:	Formerly at Emory University, relocated in September 2014 (Ed., 7/7/15)		
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Space Radiobiology NNJ08ZSA001N
Start Date:	09/20/2014	**End Date:**	03/19/2016
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Simonsen, Lisa	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: Extended to 3/19/2016 (original end date was 9/19/2015) per S. Monk/LaRC and NSSC information (Ed., 4/7/16)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX14AT39G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: Continuation of "Molecular Basis of DNA Repair and Protection from Apoptosis in Neuronal Progenitors Exposed to Space Radiation" grant NNX08BA08G, due to PI move to Prairie View A&M University in September 2014. The health risks to astronauts exposed to space radiation include cognitive deficits and possibly accelerated aging. While the pathogenesis of radiation-induced cognitive dysfunction remains largely uncharacterized, it is thought to include loss of neural progenitors from the brain. Understanding of the molecular and cellular bases underlying neuronal loss and/or dysfunction is absolutely required for the development of countermeasures before, during, and possibly after space missions. Since experiments in humans are not possible, studies in this direction will benefit from appropriate biological model systems. The neurodegenerative effects of space radiation are likely to derive from DNA damage in the central nervous system (CNS). Therefore, research involving repair of this type of DNA lesions is critical for the development of new neuroprotective countermeasures. In the present proposal, we introduce an in vitro model of neural progenitors (neurospheres), which is derived from the brain of mouse embryo from neurodegenerative transgenic mice to study the detrimental effects of space radiation at the mechanistic level. Using this biological model, we will study DNA damage repair and apoptosis of proliferating and differentiated neural progenitor exposed to low dose of high charge and energy nuclei and protons. The proposed studies will provide novel insights into the molecular and cellular mechanisms underlying CNS risks from space radiation and will help to predict and countermeasure health risks from space radiation particularly with regard to effects on the CNS.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This proposal will study the mechanism of DNA damage and oxidative stress in neuronal cells induced by high energy particle, iron, and protons, compared to X-ray. This study will provide possible ways to develop accurate quantitative estimates to the risk of the central nervous system (CNS) from galactic cosmic ray (GCR) and solar particles events following long-term space travel.
Task Progress:	In this grant period, we investigated biological effects of secondary particles from high energy proton and high energy charge particle in neuronal cells. The human health risk during space mission may have cause from solar particle events and galactic cosmic rays comprised of energy proton and high energy charge particles as well as the secondary radiation from the shielding and human tissues. The Martian regolith is one of important shielding materials during Mars exploration. The biological effects of Martian environment and the secondary radiation remain uncertain. Here we detected primary and secondary radiation tracks of high energy proton (1 GeV/n), high energy iron (1 GeV/n), and carbon (290 MeV/n) with and without Martian regolith simulant using a 1.67 uM pixel detector, and investigated the effect of secondary radiation from Martian regolith on DNA damage and cell survival in hippocamapal neuronal cells. The radiation tracks have been detected at 2 cGy to 8 cGy of proton, iron, and carbon radiation beams using this pixel detector. When Martian regolith put on the surface of a pixel detector, the secondary radiation tracks were captured by the pixel detector. Using live imaging of mCherry 53BP1 in mouse hippocampal neuronal cells, we found that DNA damage tracks are formed in cells behind Martian regolith with high energy proton, but reduced DNA damage tracks in cells behind Martian regolith with high energy iron particles. Martin regolith between proton beams and cell flasks reduced the cell survival of hippocampal neuronal cells using clonogenic assay. The secondary radiation from Martian regolith also decreased neurite outgrowth and cell viability of mouse primary hippocampal neuronal cells. Further study on the effect of secondary radiation from Martian regolith on neuronal cells is warranted.
Bibliography Type:	Description: (Last Updated: 11/13/2019)
Abstracts for Journals and Proceedings	Wang H, Hu X, Rhone JR, Holland SD, Saganti P. "Biological effects of secondary particles from high energy proton and high energy charge particle in neuronal cells." 2016 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 8-11, 2016, and the 27th Annual NASA Space Radiation Investigators' Workshop Galveston, USA, Feburary 8-11, 2016. 2016 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 8-11, 2016, and the 27th Annual NASA Space Radiation Investigators' Workshop Galveston, USA, Feburary 8-11, 2016. , Feb-2016
Articles in Peer-reviewed Journals	Burgado J, Harrell CS, Eacret D, Reddy R, Barnum CJ, Tansey MG, Miller AH, Wang H, Neigh GN. "Two weeks of predatory stress induces anxiety-like behavior with co-morbid depressive-like behavior in adult male mice." Behav Brain Res. 2014 Dec 15;275:120-5. Epub 2014 Sep 6. https://doi.org/10.1016/j.bbr.2014.08.060 ; PubMed PMID: 25200517 , Dec-2014